

The empirical recurrence for OEIS A269637 is false at every index its own data can test, and the correct one, of order 13

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### Abstract

OEIS A269637 carries an empirical recurrence of order 10 contributed by R. H. Hardin. It is false, and it is false in the strongest sense available: the entry publishes 20 terms with offset 1, so the recurrence asserts something at each of  $n = 11, \dots, 20$ , and it fails at all ten. The first failure is at  $n = 11$ , where the entry's own DATA gives 199677806 and the recurrence gives 199034092. No model and no computation beyond ten multiply-adds on the entry's published integers are needed for that. The sequence is a transfer-matrix count, so it is  $C$ -finite and its true minimal recurrence can be computed and proved; it has order 13, and its first 10 coefficients are exactly the published ones. The published line is therefore the correct recurrence with its last three terms dropped. Both statements are proved here in exact integer arithmetic.

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## 1 The sequence and the claim

OEIS A269637 is “Number of length- $n$  0..5 arrays with no repeated value differing from the previous repeated value by other than plus two or minus 1.”. It has offset 1 and its published DATA is

6, 36, 210, 1209, 6876, 38738, 216528, 1202353, 6639294, 36486190,  
199677806, 1088813760, 5918190122, 32077034918, 173421804818, 935474530883,  
5035906367464, 27059901888584, 145161762258568, 777537033208493

— 20 terms in all. The entry carries one formula line:

Empirical:  $a(n) = 29*a(n-1) - 330*a(n-2) + 1730*a(n-3) - 2815*a(n-4) - 9981*a(n-5) + 36088*a(n-6) + 31192*a(n-7) - 161403*a(n-8) - 158561*a(n-9) + 291970*a(n-10)$

As of the live entry (revision 4, last modified Mar 02 2016) this is recorded as empirical and nothing on the entry records it as settled or as corrected. Written out, the claim is

$$\begin{aligned} a(n) = & 29a(n-1) - 330a(n-2) + 1730a(n-3) - 2815a(n-4) - 9981a(n-5) \\ & + 36088a(n-6) + 31192a(n-7) - 161403a(n-8) - 158561a(n-9) + 291970a(n-10), \end{aligned} \tag{1}$$

of order 10, with no range stated and no initial index excluded.

With offset 1 and order 10, the first index at which (1) asserts anything is  $n = 11$ , and the entry publishes terms through  $n = 20$ . Unlike the usual case, the claim is therefore testable — ten times over — on the entry's own data, with no model involved.

## 2 The published recurrence fails on the published data

Substituting the entry's own integers into the right-hand side of (1) gives the following.

| $n$ | entry's $a(n)$  | (1) predicts    | difference    |
|-----|-----------------|-----------------|---------------|
| 11  | 199677806       | 199034092       | 643714        |
| 12  | 1088813760      | 1084932650      | 3881110       |
| 13  | 5918190122      | 5894918440      | 23271682      |
| 14  | 32077034918     | 31940912174     | 136122744     |
| 15  | 173421804818    | 172637008448    | 784796370     |
| 16  | 935474530883    | 931005351317    | 4469179566    |
| 17  | 5035906367464   | 5010702199870   | 25204167594   |
| 18  | 27059901888584  | 26918901378892  | 141000509692  |
| 19  | 145161762258568 | 144378240753606 | 783521504962  |
| 20  | 777537033208493 | 773207835052431 | 4329198156062 |

**Proposition 1.** *The recurrence (1) is false. It fails at every one of the ten indices  $n = 11, \dots, 20$  at which the entry's published DATA can test it, the smallest being  $n = 11$ , where the two sides differ by 643714.*

*Proof.* Direct evaluation in exact integer arithmetic, using only the twenty integers the entry publishes. For  $n = 11$ ,

$$\begin{aligned}
& 29 \cdot 36486190 - 330 \cdot 6639294 + 1730 \cdot 1202353 - 2815 \cdot 216528 - 9981 \cdot 38738 \\
& + 36088 \cdot 6876 + 31192 \cdot 1209 - 161403 \cdot 210 - 158561 \cdot 36 + 291970 \cdot 6 \\
& = 199034092 \neq 199677806 = a(11).
\end{aligned}$$

□

Nothing about  $n = 11$  is exceptional; the discrepancy grows by a factor of between 5.5 and 6 at each successive index, and the recurrence continues to fail for every larger  $n$  as well, as the corollary in Section 3 shows.

## 3 A model, and the recurrence that does hold

### 3.1 What the entry counts

Read the name as it is written. An array is a tuple  $(x_1, \dots, x_n) \in \{0, 1, 2, 3, 4, 5\}^n$ . Say that index  $i \geq 2$  *repeats* when  $x_i = x_{i-1}$ , and call  $x_i$  the repeated value there. Listing the repeated values in order of  $i$  as  $v_1, v_2, \dots$ , the condition is that each differs from its predecessor by plus two or minus one:

$$v_{j+1} - v_j \in \{+2, -1\} \quad \text{for every } j \geq 1, \quad (2)$$

with  $v_1$  unconstrained. (A run of three equal values contributes two repeated values with  $v_{j+1} - v_j = 0$ , so such runs are excluded by (2); this is forced by the data, see below.)

Let  $a(n)$  be the number of such tuples. This reading was confirmed against the entry by direct enumeration of all  $6^n$  tuples for  $n = 1, \dots, 7$ , which gives

$$6, 36, 210, 1209, 6876, 38738, 216528,$$

the entry's first seven terms exactly.

### 3.2 The transfer matrix

Whether a tuple can be extended depends on the last entry and on the last repeated value so far, and on nothing else. Take as vertex set

$$\mathcal{V} = \{0, \dots, 5\} \times (\{0, \dots, 5\} \cup \{\emptyset\}), \quad |\mathcal{V}| = S = 42,$$

a vertex  $(L, P)$  recording the last entry  $L$  and the last repeated value  $P$ , with  $\emptyset$  meaning “no repeat yet”. Define  $M$  by: from  $(L, P)$ , appending  $x \neq L$  leads to  $(x, P)$ ; appending  $x = L$  leads to  $(x, x)$  and is allowed only when  $P = \emptyset$  or  $x - P \in \{+2, -1\}$ . Let  $\mathbf{u} \in \mathbf{Z}^{\mathcal{V}}$  be the indicator of  $\{(t, \emptyset) : 0 \leq t \leq 5\}$  and  $\mathbf{1}$  the all-ones vector. Then

$$a(n) = \mathbf{1}^{\top} M^{n-1} \mathbf{u} \quad (n \geq 1). \quad (3)$$

Evaluated exactly, (3) reproduces all 20 terms of the entry’s DATA and all 210 terms of the entry’s b-file, and its first seven values agree with the brute-force enumeration above.

### 3.3 The true recurrence

Being a walk count,  $a$  is  $C$ -finite. Solving over  $\mathbf{Q}$  for a constant-coefficient recurrence from the model’s own terms, the least order that admits one is 13, and the coefficients come out integral:

$$\begin{aligned} a(n) = & 29a(n-1) - 330a(n-2) + 1730a(n-3) - 2815a(n-4) - 9981a(n-5) \\ & + 36088a(n-6) + 31192a(n-7) - 161403a(n-8) - 158561a(n-9) + 291970a(n-10) \\ & + 583350a(n-11) + 364874a(n-12) + 80630a(n-13). \end{aligned} \quad (4)$$

The first 10 coefficients of (4) are, term for term, those of (1). The published line is (4) with its last three terms,  $+583350a(n-11) + 364874a(n-12) + 80630a(n-13)$ , missing.

**Theorem 2.** (4) holds for every  $n \geq 14$ .

*Proof.* Let

$$\begin{aligned} p(t) = & t^{13} - 29t^{12} + 330t^{11} - 1730t^{10} + 2815t^9 + 9981t^8 - 36088t^7 \\ & - 31192t^6 + 161403t^5 + 158561t^4 - 291970t^3 - 583350t^2 - 364874t - 80630 \end{aligned}$$

be the characteristic polynomial of (4), and put  $w = p(M)\mathbf{u}$ . Formed by thirteen matrix–vector products in exact integer arithmetic,  $w$  is the zero vector of  $\mathbf{Z}^{42}$ . Hence for every  $n \geq 14$ ,

$$a(n) - \sum_{i=1}^{13} c_i a(n-i) = \mathbf{1}^{\top} M^{n-14} p(M)\mathbf{u} = \mathbf{1}^{\top} M^{n-14} w = 0. \quad \square$$

**Remark 3.** The annihilation here is stronger than the usual Cayley–Hamilton argument needs. Normally one checks  $\mathbf{1}^{\top} M^m w = 0$  for  $m < S$  and appeals to Cayley–Hamilton for the rest; here  $w$  itself vanishes, so  $p$  annihilates  $\mathbf{u}$  under  $M$  outright and no appeal is needed. (The weaker check was run as well, for  $m = 0, \dots, 41$ , and every value is 0.) Solving for the recurrence is only a way of *proposing* (4); what makes it a theorem is this annihilation, which is independent of how the candidate was found.

**Corollary 4.** (1) fails for every  $n \geq 11$ , not merely at the ten indices the entry’s DATA exhibits.

*Proof.* Subtracting (1) from (4), a failure of (1) at  $n$  amounts to  $583350a(n-11) + 364874a(n-12) + 80630a(n-13) \neq 0$  for  $n \geq 14$ ; every  $a(m)$  is a count and  $a(m) \geq 6 > 0$  for all  $m \geq 1$ , and all three coefficients are positive, so the left side is positive. For  $n = 11, 12, 13$  the table in Section 2 exhibits the failure directly.  $\square$

## 4 Verification

Five checks, each independent of the others.

First, the disproof in Section 2 uses nothing but the entry’s own published integers and ten multiply-adds per index. It does not depend on the model, on the b-file, or on Theorem 2.

Second, the model was not taken on trust. Its first seven values were confirmed by brute-force enumeration over all  $6^n$  tuples, it reproduces all 20 published DATA terms, and it reproduces all 210 terms of the entry’s b-file — every one, not a sample.

Third, (4) was checked directly on 260 exactly computed model terms, and holds at all 247 indices where it asserts anything, before any matrix argument was made. Orders 1 through 12 were each solved for and each fails, so 13 is minimal.

Fourth, the annihilation test of Theorem 2 was carried out in exact integer arithmetic and gave the zero vector, and the weaker Krylov test was carried to the full Cayley–Hamilton bound  $m < 42$  rather than to a sampled prefix.

Fifth, the entry’s parent table A269640 independently records the order of this column as 13: its “Empirical for column  $k$ ” block prints the recurrences for  $k = 1, 2, 3$  and then, for  $k = 4, \dots, 7$ , only the placeholders “[order 7]”, “[order 13]”, “[order 14]”, “[order 16]”. Column  $k = 5$  is A269637, and the order named there is 13, not 10.

**Remark 5.** The shape of the error decides what this note is. The published coefficients are not wrong; they are incomplete, and the line is the correct recurrence truncated after ten terms. What distinguishes this case from an over-fitted empirical recurrence reported at an order its sample cannot support is that the truncation is visible from the entry alone: the line contradicts the data printed a few lines above it, at ten consecutive indices, and the parent table names the right order. The correction is (4).

**Remark 6.** The other columns of A269640 were checked the same way. The empirical lines for A269635 ( $k = 3$ , order 7), A269636 ( $k = 4$ , order 7), A269638 ( $k = 6$ , order 14) and A269639 ( $k = 7$ , order 16) hold at every index their published DATA can test. A269643 and A269644 carry lines whose stated range “for  $n > 9$ ” and “for  $n > 10$ ” excludes the only indices at which they disagree with the data, so they too are untouched by this note. A269637 is the only member of the family whose published line is false.

## References

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